

Lecture 22: Image diffusion and multimodal generation

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain how **latent diffusion** compresses computation via a VAE to enable high-resolution image generation
2. Describe how **CLIP** creates a shared embedding space for text and images
3. Explain how **cross-attention** and **classifier-free guidance** enable text-controlled image generation
4. Compare the **U-Net** and **Transformer (DiT)** backbones for diffusion models
5. Understand **flow matching** as a simpler alternative to DDPM and trace the path to **Stable Diffusion 3**

From text to images

Bridging Lecture 21 and today

In Lecture 21, we saw how diffusion models generate **text** by iteratively unmasking tokens (discrete diffusion). Today we return to the domain where diffusion was born — **images** — and ask: how do we build systems that generate high-resolution images from text prompts?

The key challenges

Challenge	Solution (this lecture)
Images are huge (786K+ values)	Latent diffusion — compress first, diffuse in latent space
Text must control generation	CLIP + cross-attention — bind words to spatial regions
Output must match the prompt	Classifier-free guidance — amplify text alignment
Architecture must scale	DiT — replace U-Net with Transformers
Training should be simple	Flow matching — straight paths, no noise schedule

The pixel problem

Why running diffusion in pixel space is expensive

The continuous diffusion framework from Lecture 21 (DDPM) operates directly on pixel space when applied to images. For a 512×512 RGB image, the denoising network processes tensors of size $512 \times 512 \times 3 = 786,432$ values at every step. With 1000 steps, this is prohibitively expensive for high-resolution generation.

The numbers

Resolution	Pixels	U-Net memory	Time per image
64×64	12,288	~2 GB	~30 seconds
256×256	196,608	~8 GB	~5 minutes
512×512	786,432	~32 GB	~20 minutes
1024×1024	3,145,728	Not feasible	—

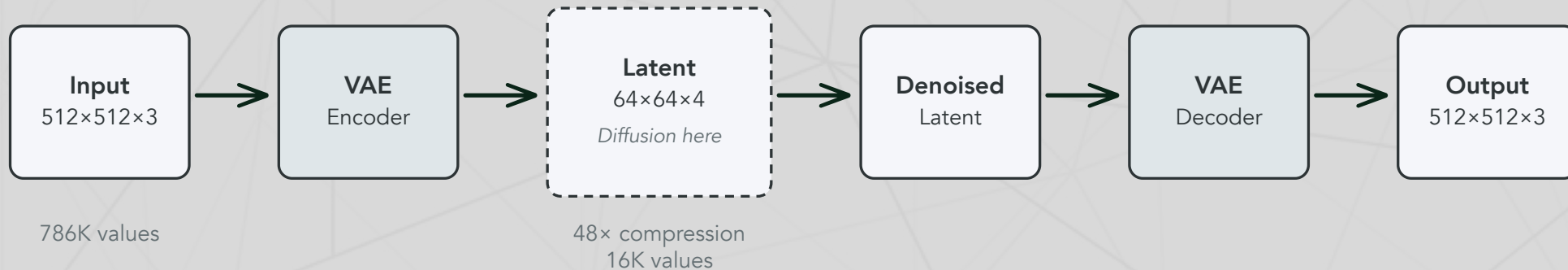
The solution: don't run diffusion in pixel space. Run it in a **compressed latent space**.

Latent diffusion models

Rombach et al. (2022, CVPR): 'High-Resolution Image Synthesis with Latent Diffusion Models'

Latent diffusion separates the problem into two stages:

1. **Compression:** A pretrained VAE (variational autoencoder) encodes images into a compact latent space (typically 8× spatial compression)
2. **Generation:** Diffusion operates entirely in this latent space



Why this works

The VAE learns to compress images while preserving perceptually important information. Diffusion in latent space is **~50× cheaper** than in pixel space, with negligible quality loss. This single insight enabled Stable Diffusion — the first open-source, consumer-GPU-capable image generator.

The VAE bottleneck

Compressing images for efficient diffusion

The VAE encoder maps an image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ to a latent $\mathbf{z} \in \mathbb{R}^{h \times w \times c}$ where $h = H/f$, $w = W/f$, and f is the downsampling factor (typically $f = 8$).

Stable Diffusion's VAE

Property	Value
Input resolution	$512 \times 512 \times 3$
Latent resolution	$64 \times 64 \times 4$
Compression ratio	$48\times$ (786K $\rightarrow$ 16K values)
VAE training	Perceptual loss + adversarial loss
Reconstruction quality	Near-lossless for natural images

The VAE is trained once and frozen. The diffusion model only ever sees latents — it never touches pixel space during training or sampling.

CLIP: understanding text for images

Radford et al. (2021, ICML): Contrastive Language-Image Pre-training

CLIP learns a **shared embedding space** for text and images using contrastive learning on 400 million image-text pairs from the internet:

1. **Image encoder** (ViT or ResNet): Maps an image to a vector
2. **Text encoder** (Transformer): Maps a caption to a vector
3. **Contrastive training**: Matching image-text pairs are pulled together; non-matching pairs are pushed apart

Why CLIP matters for diffusion

CLIP provides the "language understanding" for text-to-image systems. When you type a prompt like "a cat wearing a hat," CLIP's text encoder converts it into an embedding that captures the **visual meaning** of those words — not just their linguistic meaning. This embedding then guides the diffusion process through cross-attention.

A shared space for two modalities

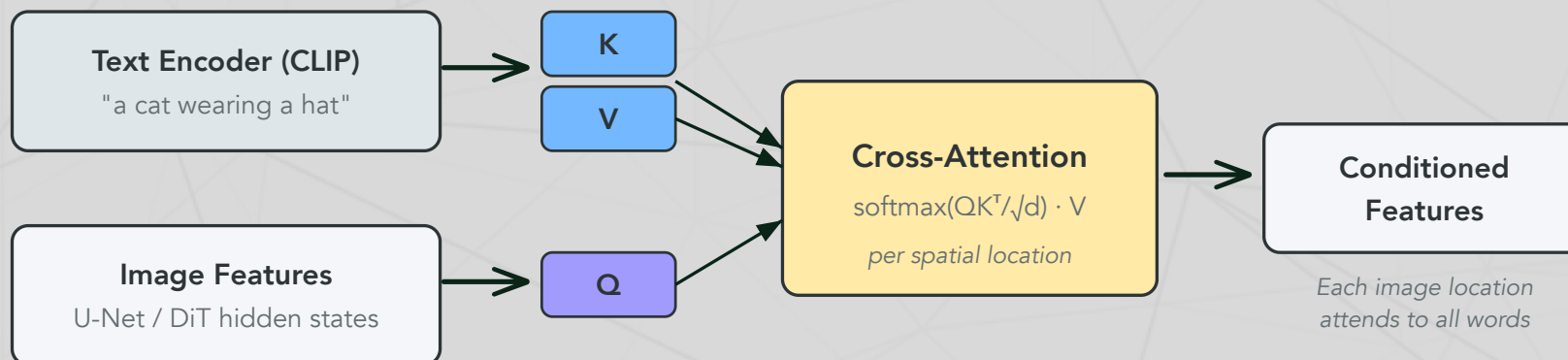
In CLIP's embedding space, the text "a golden retriever on a beach" is **near** a photo of a golden retriever on a beach, and **far** from a photo of a city skyline. This geometric

Text conditioning with cross-attention

How text controls image generation

To generate images from text prompts, latent diffusion adds **cross-attention** layers to the denoising backbone. At each spatial resolution:

1. The text prompt is encoded by CLIP into a sequence of embeddings
2. The backbone's intermediate features serve as **queries** (Q)
3. The text embeddings serve as **keys** (K) and **values** (V)
4. Cross-attention allows each spatial location in the image to attend to relevant words



Classifier-free guidance

Ho & Salimans (2022): controlling generation quality

Classifier-free guidance (CFG) is a technique for improving the alignment between text prompts and generated images. During training, the text condition is randomly dropped (replaced with an empty prompt) some fraction of the time. At inference, the model makes two predictions:

- **Conditional:** $\epsilon_{\theta}(\mathbf{x}_t, t, c)$ — with the text prompt
- **Unconditional:** $\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)$ — without the text prompt

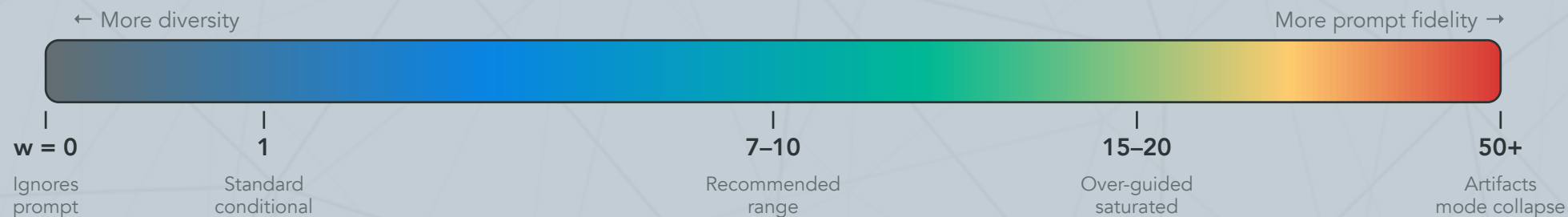
The final prediction extrapolates *away* from the unconditional toward the conditional:

$$\tilde{\epsilon}_{\theta} = (1 + w) \epsilon_{\theta}(\mathbf{x}_t, t, c) - w \epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)$$

where w is the **guidance scale** (typically 7–15).

The guidance scale tradeoff

What different guidance scales produce



The intuition

Think of CFG as asking: "What's different about images that match this prompt versus random images?" Then amplifying that difference. Higher guidance = more amplification = more faithful to the prompt but less natural variation.

In practice

Nearly every text-to-image system uses CFG. Stable Diffusion defaults to $w = 7.5$; DALL-E 2 uses $w \approx 4$. The "recommended range" of 7-10 balances prompt fidelity against visual quality — pushing higher creates oversaturated, artifact-prone images.

U-Net: the original diffusion backbone

Ronneberger et al. (2015, MICCAI): U-shaped architecture

The U-Net was originally designed for biomedical image segmentation, but became the default backbone for diffusion models from DDPM through Stable Diffusion 1–2:

1. **Encoder path** (downsampling): Progressively reduces spatial resolution while increasing channels — captures global context
2. **Decoder path** (upsampling): Progressively restores spatial resolution — generates fine details
3. **Skip connections**: Connect encoder layers to decoder layers at matching resolutions — preserve high-frequency spatial information

U-Net in diffusion models

Property	Typical configuration
Input/output	Same-sized latent (e.g., $64 \times 64 \times 4$)
Conditioning	Timestep t via sinusoidal embeddings; text via cross-attention
Skip connections	Concatenate encoder features to decoder features
Parameters	~860M (Stable Diffusion 1.5)

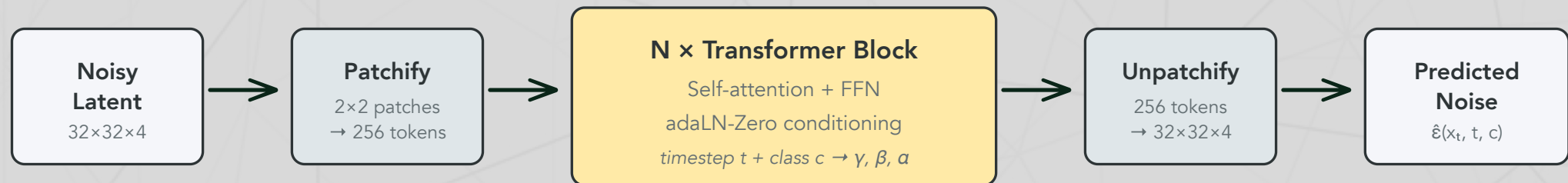
The U-shaped design gives diffusion models **multi-scale reasoning**: coarse structure from the bottleneck, fine detail from skip connections.

Diffusion Transformer (DiT)

Peebles & Xie (2023, ICCV): replacing U-Net with Transformer

The Diffusion Transformer (DiT) replaces the U-Net backbone with a standard Vision Transformer (ViT):

1. **Patchify**: Divide the latent into non-overlapping patches (e.g., 2×2)
2. **Flatten**: Treat patches as a sequence of tokens (just like ViT treats image patches)
3. **Process**: Apply standard Transformer blocks with self-attention
4. **Unpatchify**: Reshape back to spatial dimensions



Treats latent patches as tokens — standard ViT architecture

Why replace U-Net?

Transformers scale better than U-Nets. DiT-XL/2 (675M parameters) achieves a new state-of-the-art FID of 2.27 on ImageNet, beating all previous diffusion models. More importantly, DiT shows **clean scaling behavior** — larger models consistently produce better results, with no architectural bottlenecks.

adaLN-Zero: conditioning in DiT

Adaptive layer normalization for timestep and class conditioning

DiT conditions on timestep and class label using **adaptive Layer Normalization (adaLN-Zero)**:

1. The timestep and class embeddings are combined and projected to produce **scale** (γ) and **shift** (β) parameters
2. These modulate the LayerNorm output: $\text{adaLN}(\mathbf{h}) = \gamma \odot \text{LN}(\mathbf{h}) + \beta$
3. Additionally, a **gating parameter** α scales the residual connection, initialized to zero

Why 'Zero'?

Initializing the gating parameter $\alpha = 0$ means each Transformer block initially acts as an **identity function**. This makes training stable even for very deep models — the network starts by doing nothing and gradually learns to denoise. This is the same principle behind residual learning ([He et al., 2016](#)).

Flow matching

Lipman et al. (2023, ICLR): a simpler framework

Flow matching offers a cleaner mathematical framework than DDPM. Instead of a discrete chain of noising steps, flow matching defines a **continuous path** from noise to data using an ordinary differential equation (ODE):

$$\frac{d\mathbf{x}}{dt} = v_{\theta}(\mathbf{x}_t, t)$$

where v_{θ} is a neural network that predicts the **velocity** (direction and speed) of the flow at each point.

Remember...

- **ODE** (Ordinary Differential Equation): An equation describing how a quantity changes over time via a deterministic rule — given the current state, the next state is fully determined
- **SDE** (Stochastic Differential Equation): Like an ODE but with a random noise term — the path from noise to data has some randomness at each step

Flow matching vs DDPM

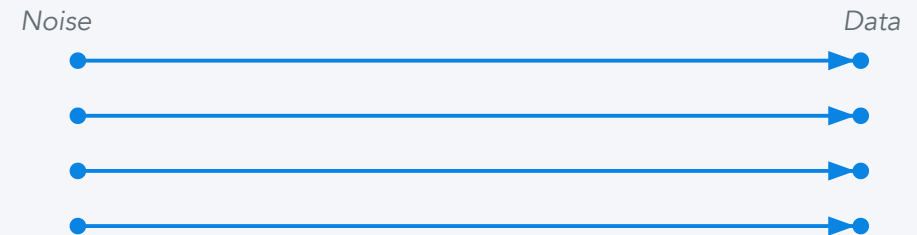
Key differences

	DDPM	Flow matching
Path type	Stochastic (SDE)	Deterministic (ODE)
Training	Predict noise ϵ	Predict velocity v
Interpolation	$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$	$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \epsilon$
Simplicity	Requires noise schedule design	Schedule-free

DDPM (curved paths)



Rectified flow (straight paths)



The intuition: Flow matching defines a straight interpolation from noise to data and learns the velocity field along it — no noise schedule needed.

Rectified flow

Straight paths from noise to data

Rectified flow (Liu et al., 2023, ICLR) uses the simplest possible interpolation — a straight line between the data point and a noise sample:

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \epsilon$$

The velocity along this path is constant: $v = \epsilon - \mathbf{x}_0$

Why straight paths matter

Straight paths are the shortest paths between noise and data. Since they don't curve, they can be accurately simulated with **fewer ODE solver steps** — as few as 1–4 steps with distillation. This is the foundation of Stable Diffusion 3's fast generation.

Stable Diffusion 3: putting it all together

Esser et al. (2024, ICML): MMDiT architecture

Stable Diffusion 3 combines all of the techniques we've discussed into a single system:

Component	Choice	Why
Latent space	VAE with 16-channel latent	Higher capacity than SD 1.x (4 channels)
Backbone	MMDiT (multimodal DiT)	Transformers scale better than U-Net
Text encoders	CLIP-L + CLIP-G + T5-XXL	Three encoders for rich text understanding
Conditioning	Joint attention (text + image tokens)	Text and image tokens attend to each other
Training framework	Rectified flow	Simpler than DDPM, faster sampling
Guidance	CFG with dynamic rescaling	Better prompt adherence

The trend

Each generation of image diffusion models combines insights from the previous one. SD3 isn't one breakthrough — it's the accumulation of latent diffusion + CLIP + CFG + DiT + flow matching + better text encoders. Understanding the individual pieces (this lecture) is the key to understanding the whole.

Discussion

Questions to consider

1. **Latent vs pixel space:** Latent diffusion trades exact pixel control for speed. Are there tasks where pixel-space diffusion would be strictly better? What information might the VAE discard?
2. **Guidance as amplification:** CFG amplifies the difference between conditional and unconditional predictions. Is this analogous to anything in human cognition — e.g., how we exaggerate features when imagining something vividly?
3. **U-Net vs Transformer:** DiT replaced U-Net because Transformers scale better. But U-Net's inductive biases (locality, skip connections) seem useful for spatial data. Is there a hybrid that gets the best of both?
4. **Language drives vision:** Imagen (Lecture 23) found that scaling the text encoder improves image quality more than scaling the diffusion model. What does this tell us about the relationship between language and visual imagination?

Take-home messages

Think about it...

- The key bottleneck in high-resolution generation wasn't the diffusion process itself — it was **where** you run it. Compressing to latent space (via a VAE) made consumer-GPU generation possible.
- Text-to-image generation requires a **bridge between modalities**: CLIP creates a shared embedding space, and cross-attention lets the diffusion model "listen" to text at every spatial location.
- Classifier-free guidance shows that **controlling** generation is as important as generation itself — and the trick is surprisingly simple: learn what the conditional and unconditional outputs look like, then amplify the difference.
- The field evolves by **composing** innovations (latent space + CLIP + CFG + DiT + flow matching), not replacing them. Each addresses one specific limitation.

Further reading

Further reading

Rombach et al. (2022, CVPR) "High-Resolution Image Synthesis with Latent Diffusion Models" — Latent diffusion and the foundation of Stable Diffusion.

Radford et al. (2021, ICML) "Learning Transferable Visual Models From Natural Language Supervision" — CLIP: the shared embedding space that makes text-to-image generation possible.

Ho & Salimans (2022, arXiv) "Classifier-Free Diffusion Guidance" — The guidance technique used in virtually all modern diffusion systems.

Ronneberger et al. (2015, MICCAI) "U-Net: Convolutional Networks for Biomedical Image Segmentation" — The U-shaped architecture that served as the default diffusion backbone.

Peebles & Xie (2023, ICCV) "Scalable Diffusion Models with Transformers" — DiT: replacing U-Net with Transformers for clean scaling.

Lipman et al. (2023, ICLR) "Flow Matching for Generative Modeling" — A simpler, ODE-based alternative to diffusion SDEs.

Esser et al. (2024, ICML) "Scaling Rectified Flow Transformers for High-Resolution Image Synthesis" — Stable Diffusion 3 and the MMDiT architecture.

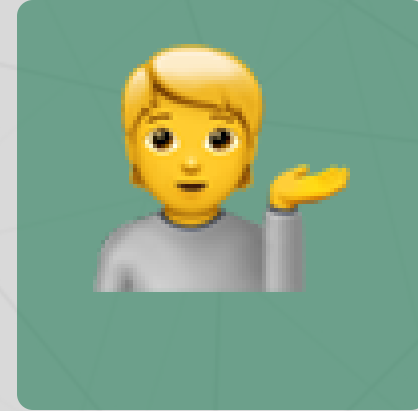
Questions?



Email



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Office hours

Up next...

Diffusion applications: text-to-image, text-to-video, and the ethics of generative AI